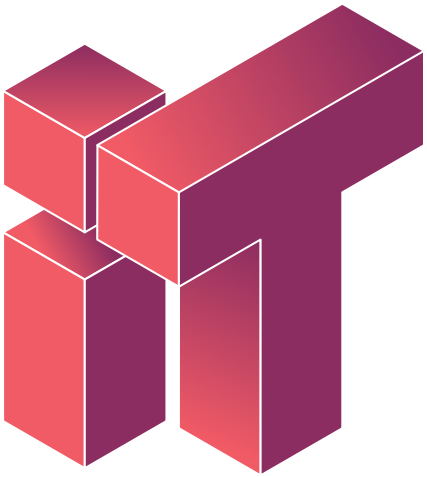
Data Parallelism with Rayon

Laboratory - Advanced Programming



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1 Goal

Modern CPUs have multiple cores, but most programs only use one. **Data parallelism** lets you harness all cores by processing chunks of data simultaneously.

In this laboratory you will refactor the Humans vs. Zombies simulation to use Rayon, Rust's data-parallelism library. You will learn to:

- Identify loops that can safely run in parallel
- Transform sequential iterators into parallel ones
- Recognise when a loop **cannot** be parallelised due to sequential dependencies
- Measure the speedup gained from parallel execution

1.1 What is Rayon?

Rayon turns sequential iterator chains into parallel ones. The API mirrors the standard iterator methods — you only need to add **`par_`**:

```
// Sequential
let sum: u32 = numbers.iter().map(|n| n * n).sum();

// Parallel (just add par_)
let sum: u32 = numbers.par_iter().map(|n| n * n).sum();
```

Rayon automatically splits your data across available CPU threads, processes chunks in parallel, and collects the results transparently.

The key requirement is that the data type must be **`Send`** (safe to transfer between threads) and **`Sync`** (safe to share via references). You will add these bounds to the **`Entity`** trait.

1.2 Setup

Clone the repository:

```
git clone <repo-url>
cd swd-course
```

2 Refactor with Rayon

2.1 Baseline

```
cargo run --release -- bench
```

- Run **`cargo run --release -- bench`**.
- Note the averages – you will fill them in Table 1 at the end of the document.

2.2 Add Rayon to the Project

```
cargo add rayon
```

- Add **`rayon`** to **`Cargo.toml`**.
- Add **`Send + Sync`** to the **`Entity`** trait in **`entity.rs`**.
- Add **`use rayon::prelude::*;`** in **`world.rs`**.
- **`cargo test`** passes.

2.3 count_humans / count_zombies

Refactor both methods to parallelise them.

- Parallelise both methods.
- **`cargo test`** passes.

2.4 tick_decay

Refactor the **`for`** loop to parallelise it.

- Parallelise **`tick_decay`**.
- **`cargo test`** passes.

2.5 tick_infection

Parallelise the check phase (read-only). The removal and creation loops must stay sequential.

- Parallelise the infection check phase.
- **`cargo test`** passes.

2.6 tick_movement

Can **`tick_movement`** be parallelised with Rayon? Why or why not? What would happen if two entities computed their positions simultaneously?

- Understand why **`tick_movement`** stays sequential.
- All **`cargo test`** tests pass.

2.7 Benchmark

Run **`bench`** on your refactored version:

```
cargo run --release -- bench
```

Fill in your results and compare with your baseline:

Phase	Baseline (sequential)	Refactored (Rayon)	Δ
movement			
infection			
decay			
total			

Table 1 - Benchmark comparison (fill in your measurements)

Compare your measurements and answer:

1. Which phase benefits **most** from Rayon? Why?
2. Which phase gets **slower** with Rayon? Why?
3. Why doesn't movement change between the two versions?
4. What does this tell you about when parallelism is worth it?

3 Checkout

Before finishing, ensure you have completed:

- [ ] Setup
  - [ ] **`rayon`** added to **`Cargo.toml`**
  - [ ] **`Send + Sync`** added to the **`Entity`** trait
  - [ ] **`use rayon::prelude::*;`** in **`world.rs`**
- [ ] Counting (**`count_humans`** / **`count_zombies`**)
  - [ ] Parallelised with Rayon
  - [ ] Tests pass
- [ ] Decay (**`tick_decay`**)
  - [ ] Parallelised with Rayon
  - [ ] Tests pass
- [ ] Infection (**`tick_infection`**)
  - [ ] Check phase parallelised with Rayon
  - [ ] Removal and creation loops stay sequential
  - [ ] Tests pass
- [ ] Movement (**`tick_movement`**)
  - [ ] Understand why it stays sequential
  - [ ] Tests pass
- [ ] Benchmark
  - [ ] Baseline recorded at start
  - [ ] Refactored version benchmarked
  - [ ] Table filled and analysis questions answered
- [ ] Final verification
  - [ ] **`cargo test`** reports all tests passing
  - [ ] **`cargo run`** shows the game running correctly

Glossary